Aditya Narayan Jha

adityajha8906@gmail.com | +91 9153831641 | LinkedIn | GitHub

SUMMARY

AI Engineer with hands-on experience in LLM fine-tuning, RAG, multi-agent systems and production ML pipelines. Built and pretrained a 125M-parameter text-to-SQL language model from scratch.

SKILLS

- **Languages:** Python, SQL.
- **AI & Machine Learning:** LLMs, Fine-tuning, QLoRA, Pretraining, Text-to-SQL, RAG, LangChain, LangGraph, LlamaIndex, Multi-Agent Systems, MCP, Transformers, Hugging Face, PyTorch, TensorFlow, Scikit-Learn, Machine Learning, Clustering.
- **Data Engineering:** ETL Pipelines, Airflow, Vector Databases, Neo4j, PostgreSQL, MongoDB.
- **Backend & Cloud:** FastAPI, Node.js, REST APIs, GCP, GitHub Actions, Docker, Git.

EXPERIENCE

AI Engineer Sep 2025 – Present
Coherent Market Insights Pune, India

CoherentConnect – AI B2B buyer discovery and outreach platform

- Designed a **trend-discovery engine**: embedded news articles with **text-embedding-3-small** and constructed a 4-level trend hierarchy via **recursive K-Means** (silhouette-based splits, **Kneedle** elbow detection) and **LDA** topic modeling, with **TF-IDF keywords** refined by an **LLM** into human-readable trend labels.
- Engineered the buyer-matching layer: a **LangGraph** agent maps each trend to directly and indirectly affected industries, and companies are ranked by **cosine similarity with recency decay**, surfacing **800–1,000 potential buyers/day** from **66,000+ daily signals**.

Self-Hosted LLM Systems

- Fine-tuned **Qwen2.5-7B-Instruct** with **QLoRA** (4-bit, rank 32, attention + MLP layers) on **559 production prompt–output pairs** distilled from DeepSeek, replacing the paid API behind a newsletter engine personalized for **500,000+ recipients/month**; reduced eval loss from **3.10 to 0.36** and beat the teacher on **15/20 held-out prompts** with **100% valid JSON**, for under **$5** on a single L4 GPU.
- Developed a local CRM intelligence engine: indexed **97K leads** and rep notes with nomic-embed into **110K-vector memory** served by **hybrid RAG** (vector + keyword search fused via RRF), derived a **13x60 loss-reason taxonomy** via rules and K-Means sub-clustering, and fit a **lead scorer** (logistic regression, **0.674 AUC**, outperforming gradient boosting and random forest) whose top 20% held **41.8% of wins**; a **Qwen2.5-7B router** (**92.6% accuracy**) lets **AI agents** help reps prioritize leads, diagnose lost deals and query data in plain English.

Data Science Intern Jun 2025 – Sep 2025
SparkAI.tech Remote, US

- Architected an end-to-end **ETL pipeline** (Pecan Street, OEDI datasets to PostgreSQL) with schema validation and **Isolation Forest** anomaly detection, powering ML training across **10+ device categories** and cutting ingestion failures by **45%**.
- Trained **Temporal Fusion Transformer (TFT)** models on 15-minute granularity consumption data, achieving a **26% accuracy improvement**, and applied **HDBSCAN** clustering to segment load profiles for downstream feature engineering.

PROJECTS

TinySQL – 125M-Parameter Text-to-SQL LLM Built from Scratch GitHub | Paper (DOI)

- Hand-wrote a **125M-parameter Llama-style LLM** in PyTorch: a 32K **byte-level BPE tokenizer**, the transformer (**GQA, RoPE, SwiGLU, RMSNorm**), AdamW and a resumable bf16 training loop, verified by **169 tests** (logits match HF Llama to 10^{-4}). Built a **5.4B-token corpus** with MinHash-LSH deduplication and benchmark decontamination (0 leaks in 4.7M documents), then pretrained on **4.5B tokens** in **46 hours on one L4 GPU** at 27K tokens/s.
- Fine-tuned on WikiSQL and added a **grammar-constrained decoder** (only real columns, values copied from the question), raising test execution accuracy from **77.9% to 80.4%** with **100% valid SQL**. Benchmarked against GPT-2, Pythia-160M and SmolLM2 with identical recipes, 3 seeds and McNemar significance tests: within 1 point of GPT-2 and **17 points above Pythia-160M** on 67× less data; ablations show RoPE (+8.5 pts) and tied embeddings (+5 pts) matter most. Released the paper on Zenodo and the models on Hugging Face.

EDUCATION

B.Tech in Computer Science and Engineering (Health Informatics) Sep 2022 – May 2026
Vellore Institute of Technology Bhopal, India
CGPA: 9.11/10